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1D Convolutional Autoencoder for ECG Signal Denoising and Classification

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ABSTRACT

INDEX TERMS CAE, Classification, CNN, CVD, Decoder, Denoising, ECG, Encoder, SWT

I. INTRODUCTION

Cardiovascular disease (CVD) is one of the leading causes of death globally, and it involves all disorders that affect the heart and blood circulation or vessels [1]. This is why it is important to enhance all detection techniques to be able to discover early any irregularities to avoid fatal stages. One of the most important techniques is electrocardiogram (ECG) monitoring. The Electrocardiograph is the tool used to output ECG, which is composed of a group of sensors or electrodes attached to the human skin to measure the electrical signals the heart produces with each heartbeat. Based on the output, which is the ECG signal, specialists can find out if there are any abnormalities.

There are also portable, easy-to-use ECG monitoring devices available in the market, but using them may yield inaccurate results due to noise resulted from the inability to be completely still, as the noise may distort the heart's signal, which may lead to misdiagnosis. There are many reasons for having noise during measurement.

The following are the main causes of noise:

- Noises caused by electrode movement (EM), which usually have a large amplitude.
- High-Frequency Noises caused by muscle artifact (MA), which is a muscle noise caused by involuntary muscle movements like shivering, tension, or tremors.
- Low-Frequency Noises caused by baseline wander (BW), mainly caused by human breathing.

II. LITERATURE REVIEW

III. METHODOLOGY

IV. EVALUATION

V. ABLATION STUDIES

VI. CONCLUSION

VII. CHALLENGES AND FUTURE DIRECTIONS

REFERENCES

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